XPP Studio

An introduction and tutorial

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Abstract

XPP Studio is a browser-based environment for exploring dynamical systems, modelled on Bard Ermentrout's XPPAUT. It reads XPPAUT's `.ode` files and provides phase planes, three-dimensional views, time courses, numerical continuation of equilibria and periodic orbits, one- and two-parameter bifurcation diagrams, maps, stochastic and Markov models, stiff solvers, delay equations, events (global flags), arrays and Volterra integral equations. Everything runs locally in a web page: there is nothing to install and no data leaves the browser. This document introduces the interface, describes the model language, and works through a sequence of short tutorials using the examples that come with the program.

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1 Introduction

For more than thirty years XPPAUT [1] has been one of the standard tools for simulating and analysing differential equations in mathematical biology, neuroscience and applied dynamical systems. Its strengths are a compact, readable model language, fast interactive phase-plane analysis, and a tight connection to the continuation package AUTO [2]. XPP Studio keeps that model language and that way of working, and puts them in a modern interface:

- **Same files.** Most XPPAUT `.ode` files load unchanged. Supported features include parameters, functions, fixed quantities, auxiliary variables, maps, Wiener noise, Markov variables, tables, periodic (folded) variables, arrays, Volterra integrals, delays and global flags (Section 3).
- **Live analysis.** Dragging a parameter slider immediately recomputes trajectories, nullclines and equilibria.
- **Built-in continuation.** Equilibria, periodic orbits and their bifurcations can be continued in one or two parameters without leaving the program. Detected points include folds, Hopf, period-doubling, torus, fold-of-cycles and codimension-two points.
- **Publication output.** Every view can be exported as a vector SVG figure. Continuation data can be exported as CSV, and trajectories copied as CSV.

The program is a single web page. To use it, open

`https://xpp-studio.github.io/xpp-studio/`

in any recent browser. The source code, the example models and this tutorial are in the repository `https://github.com/xpp-studio/xpp-studio`. Its file `index.html` is the whole program, and it also runs when opened directly from disk.

How this document is organised. Section 2 is a tour of the interface and Section 3 summarises the model language. Sections 4–11 are hands-on tutorials, each built on an example from the **Model** menu. Section 14 covers numerical settings and Section 15 exporting results. The document ends with the differences from XPPAUT and with credits.

2 A tour of the interface

The window (Figure 1) has three columns.

Left: the model. Four tabs:

Equations The `.ode` text, which you can edit freely. **Apply** (or `Ctrl`+`Enter`) reloads the model, and messages underneath report what was loaded and any errors, with line numbers. **Open .ode** loads a file from disk, **Copy text** copies the current text, **Write state** writes the current parameters, initial conditions and settings into it (Section 15), and **Table files** supplies data files used by `table` declarations.

Parameters One row per parameter, with a value box and a slider whose limits you can edit. Drag a slider to see everything update live, and use **Reset** to restore the values in the file. The up and down arrows on a value box, and the arrow keys on a slider, change the value by 1% of the slider's range (10% with `Shift`).

Initial Initial values of the state variables (and Markov states). **Integrate** runs from here, and **Last point** → **IC** copies the end of the current trajectory into the initial conditions.

Numerics Integration method, total time, transient, step size, tolerance and bound, plus the continuation settings (Section 14).

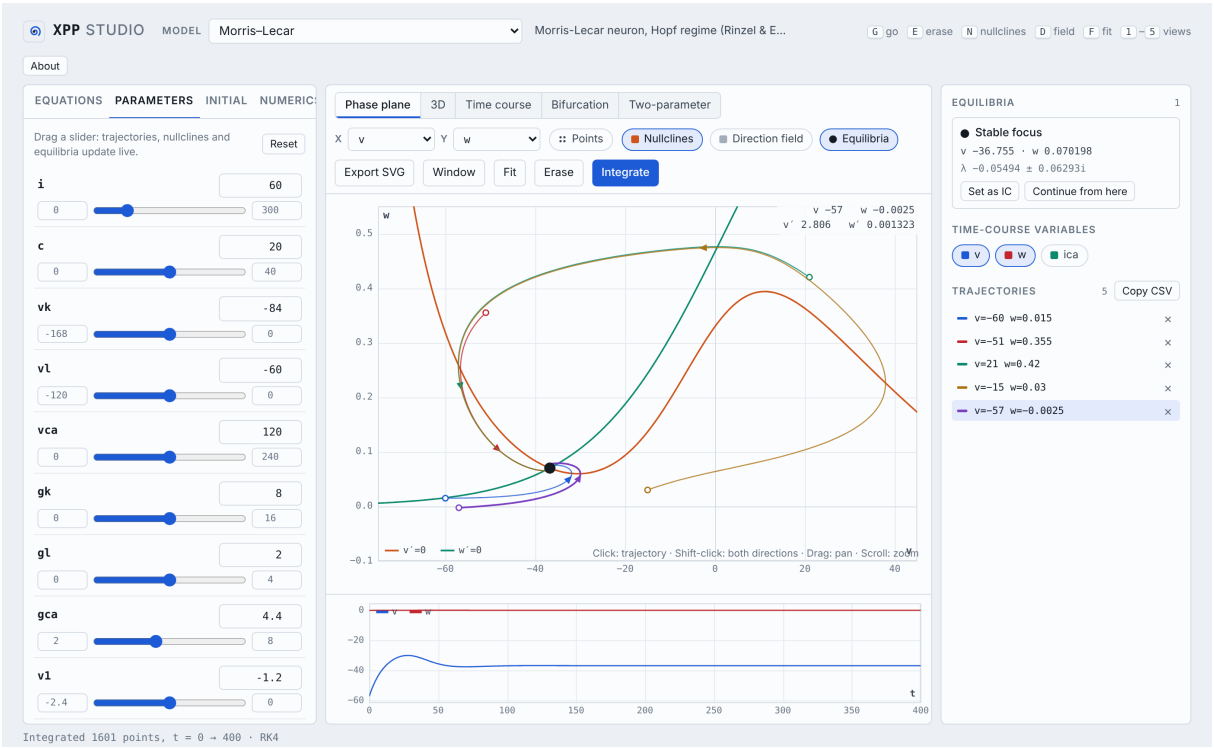


Figure 1: The XPP Studio window with the Morris–Lecar example loaded. Left: tabs for equations, parameters, initial conditions and numerics. Centre: the plot, here the (v, w) phase plane with nullclines, five trajectories and a stable focus, plus a time-course strip underneath. Right: the inspector, listing equilibria, the variables shown in the time course, and the trajectories.

Centre: the plot. Five views, chosen with the buttons above the plot or with the keys 1–5:

Phase plane Any two variables (state, Markov, Volterra or auxiliary) against each other, with optional nullclines, direction field and equilibria. For a one-dimensional map this view shows a cobweb diagram. **Equal axes** gives both axes the same scale. Below the plane a strip shows either the time series of the current trajectory or, with **Array**, an array plot (Section 11).

3D Three variables (or time) in a rotatable box: drag to rotate, scroll to zoom, **Spin** to animate.

Time course Chosen variables against t (or the iteration number n for maps).

Bifurcation Branches of equilibria and periodic orbits, and parameter sweeps, against one parameter. The vertical axis is a variable, or the period or frequency of the periodic orbits.

Two-parameter Curves of folds, Hopf points, period doublings and bifurcations of cycles in a parameter plane.

In every view you drag to pan and scroll to zoom. **Fit** rescales to the data, and **Window** sets the axis limits by hand (they then stay fixed until you press **Fit**). **Points** draws trajectories as dots instead of lines, which is the default for maps. **Export SVG** saves the current view as a vector figure. **Export CSV** saves the data behind the time course, bifurcation diagram or two-parameter diagram, and **Export .dat** saves continuation data as plain columns of numbers (Section 15).

Right: the inspector. The inspector's contents depend on the view. In the phase plane it lists the equilibria (type, coordinates and eigenvalues, with buttons to make one the initial condition, start continuation from it, or draw its separatrices), the variables in the time course, and the trajectories. In the Bifurcation view it holds the continuation controls and the table of special points. In the Two-parameter view it holds the two-parameter controls.

Key	Action
G	integrate from the current initial condition (“go”)
E	erase all trajectories
N / D	toggle nullclines / direction field
F	fit the axes to the data
1–5	phase plane, 3D, time course, bifurcation, two-parameter
Esc	during continuation: skip to the other direction (“soft interrupt”)
Shift+Esc	stop the continuation altogether
click / shift-click	start a trajectory (shift: forwards and backwards in time)

Table 1: Keyboard shortcuts and mouse actions.

3 The model language

A model is a plain-text `.ode` file in XPPAUT’s format. Lines starting with `#` are comments, and the first comment line becomes the model’s title. Names are case-insensitive. The Morris–Lecar example [9, 10] illustrates most of the basic syntax:

```
# Morris-Lecar neuron, Hopf regime (Rinzel & Ermentrout)
dv/dt=(i+gl*(vl-v)+gk*w*(vk-v)+gca*minf(v)*(vca-v))/c
dw/dt=phi*(winf(v)-w)/tauw(v)
minf(v)=.5*(1+tanh((v-v1)/v2))
winf(v)=.5*(1+tanh((v-v3)/v4))
tauw(v)=1/cosh((v-v3)/(2*v4))
aux ica=gca*minf(v)*(v-vca)
param i=60,c=20
param vk=-84,vl=-60,vca=120
param gk=8,gl=2,gca=4.4
param v1=-1.2,v2=18,v3=2,v4=30,phi=.04
init v=-60,w=0.015
#@range i 0 300
@ total=400,dt=0.25,meth=rk4,xp=v,yp=w
@ xlo=-75,xhi=45,ylo=-0.1,yhi=0.55
done
```

Table 2 summarises the language. The `#@range` line is an XPP Studio extension that sets a slider’s range. Because it starts with `#`, XPPAUT reads it as a comment, so files stay portable.

Methods. `meth=` accepts `rk4` (fixed-step Runge–Kutta), `euler`, `dp45` (adaptive Dormand–Prince 4(5)), and the stiff names `stiff`, `gear`, `cvode`, `rosenbrock` and `backeul`. All of the stiff names select an adaptive Rosenbrock 2(3) solver [7]. `discrete` iterates a map. Models with Wiener variables always use Euler–Maruyama, and models with Volterra integrals use a product-trapezoid scheme.

Syntax	Meaning
`x'=...  or  dx/dt=...`	differential equation
`x(t+1)=...`	map (or use `x'=...` with `@ meth=discrete`)
`x(0)=1  or  init x=1,y=0`	initial values
`par a=1, b=2`	parameters (also `param`, `p`)
`number c=3`	named constant
`!d=a*b`	derived parameter, recomputed when parameters change
`f(x,y)=...`	user function
`q=...`	fixed quantity, evaluated before the equations
`aux s=...`	auxiliary output, plotted like a variable
`wiener w`	white noise for an SDE, e.g. `x'=-x+s*w`
`markov z 2  {0} {kon}  {koff} {0}`	two-state Markov variable; entry (i, j) is the rate from i to j
`table h % 101 0 6.283 sin(t)`	table sampled from an expression
`table h data.tab`	table read from a file (add it with **Table files**)
`x[1..20]'=-x[j]+x[j-1]`	array: the line is repeated for $j = 1, \ldots, 20$
`%[1..20] ... %`	repeat a block of lines
`int{K(t,t',x)}`	Volterra integral $\int_0^t K(t, t', x(t'))\, dt'$
`int{K#x}`	convolution $\int_0^t K(t - t')\, x(t')\, dt'$
`int[mu]{...}`	weakly singular kernel, extra factor $(t - t')^{-\mu}$
`u(t)=...int{...}`	integral equation (also `volterra u=...`)
`delay(x,tau)`	delayed value $x(t - \tau)$; τ may depend on the state
`x(0)=0.5+0.2*cos(t)`	history of a delay equation for $t < 0$ (its value at $t = 0$ is the initial condition)
`global 1 x-1 {x=0;y=y+1}`	global flag: when `x-1` crosses zero upwards (1), downwards (-1) or either way (0), make the assignments; `arret=1` stops the run
`#@range p lo hi`	slider range for parameter `p`
`@ total, trans, dt, meth, toler, bound`	integration: duration, transient, step, method, tolerance, bound
`@ xp, yp, zp, axes=3, xlo, xhi, ylo, yhi`	initial plot variables and window
`@ fold=th, tor_per=6.2832`	periodic variable `th` with the given period
`@ seed=7`	fix the random seed
`@ vmaxpts=4000`	memory length for Volterra integrals
`@ ds, dsmin, dsmax, nmax, ntst, parmin, parmax`	continuation numerics, as in XPPAUT's AUTO window
Functions: `sin cos tan exp ln log sqrt abs heav sign max min mod atan2 sinh cosh tanh, if(c)then(a)else(b)`, ...	

Table 2: The `.ode` language understood by XPP Studio.

4 Tutorial 1: phase-plane analysis

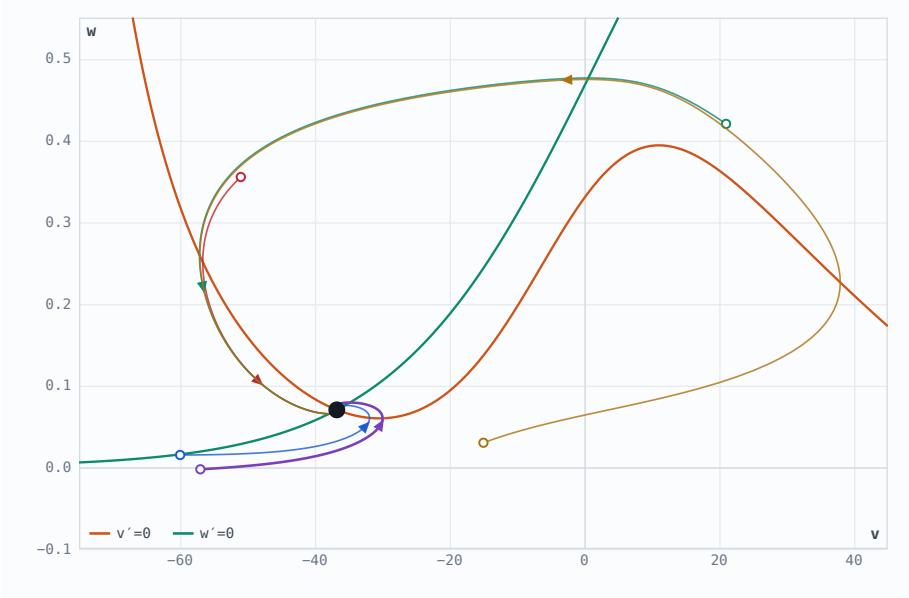


Figure 2: Morris–Lecar phase plane at $i = 60$: the v-nullcline (orange), the w-nullcline (green), a stable focus (filled circle) and several trajectories. Open circles mark starting points.

Choose **Morris–Lecar** from the **Model** menu. The phase plane shows w against the membrane potential v, together with the two nullclines.

Try it

1. Click anywhere in the plane to start a trajectory there. Shift-click integrates both forwards and backwards. Each new trajectory gets its own colour and appears in the **Trajectories** list, where you can remove it with ×.
2. Press D for the direction field and N to hide or show the nullclines. Hovering over the plot shows the coordinates and the derivatives v', w' at the pointer.
3. The inspector lists one equilibrium, a *stable focus*, with its eigenvalues $\lambda \approx -0.055 \pm 0.063i$. **Set as IC** moves the initial condition onto it.
4. Open the **Parameters** tab and slowly drag the slider for the applied current i. Between about $i = 94$ and $i = 212$ the focus becomes unstable and trajectories settle onto a large limit cycle: repetitive firing.
5. Press 3 for the time course. The chips in the inspector choose which variables are plotted, auxiliary ones included (`ica` here).

Equilibria are found by Newton's method from a grid of starting points covering the visible window, so panning or zooming out searches a wider region. For a saddle, the **Separatrices** button integrates its stable and unstable manifolds. Once drawn, they follow the saddle: when you change a parameter, the saddle is found again and its separatrices are recomputed from its new position, so you can watch, for example, a separatrix approach a homoclinic connection as you drag a slider.

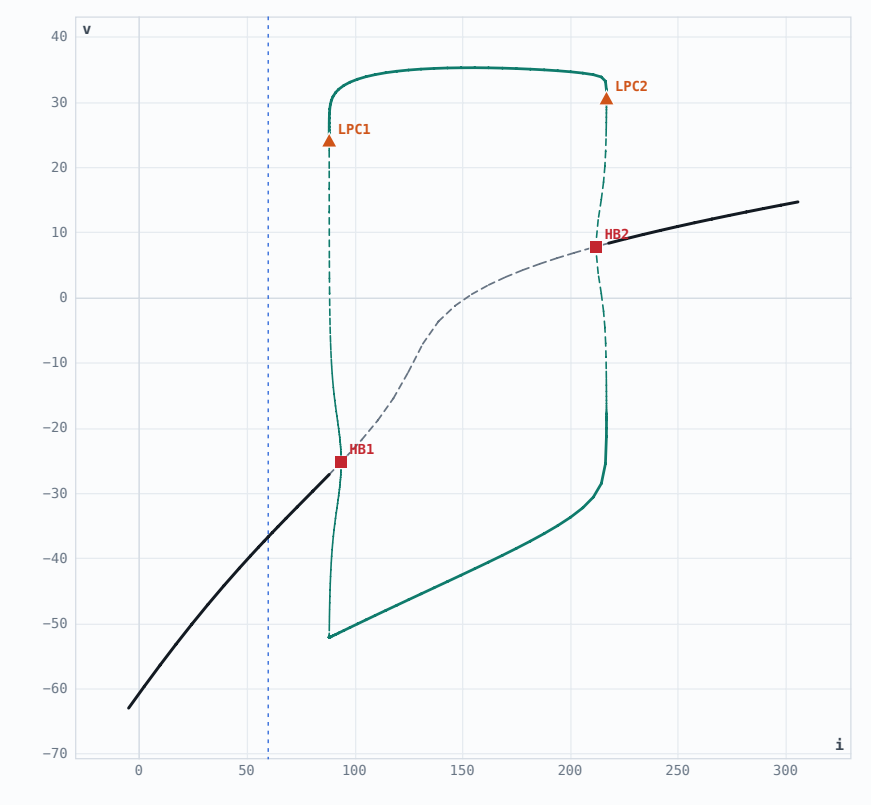


Figure 3: Bifurcation diagram of Morris–Lecar in the current i. Black: equilibria (solid stable, dashed unstable). Green: maximum and minimum of v along the periodic orbits born at the two Hopf points HB1 and HB2. Both Hopf bifurcations are subcritical. The orbit family turns at two folds of cycles, LPC1 and LPC2, giving bistability between rest and firing.

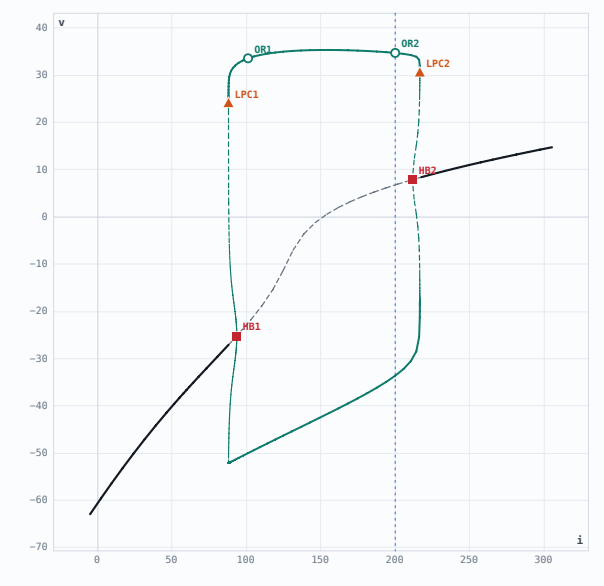
5 Tutorial 2: one-parameter bifurcation diagrams

Press 4 for the Bifurcation view. The inspector shows the continuation controls: the **Parameter** to vary, the variable to **Plot**, the parameter **Range**, the initial step **Step ds**, and the equilibrium to **Start** from.

Try it

1. Press **Continue equilibria**. Pseudo-arclength continuation [6] traces the branch in both directions across the range. Stability is read off the eigenvalues, and two Hopf points appear: HB1 at $i \approx 93.68$ (period ≈ 79.2) and HB2 at $i \approx 212.06$. Each Hopf point is tagged *sub* or *super* in the **Special points** table (Figure 6). The tag comes from the sign of the first Lyapunov coefficient ℓ_1, and hovering over it shows the value. Both points here are subcritical ($\ell_1 > 0$), so the orbits born there are unstable.
2. In the **Special points** table, press **Cycles** on the row for HB1. The periodic orbits born there are computed by multiple shooting with an integral phase condition, as in AUTO, and their stability comes from Floquet multipliers. While the computation runs, Esc (or **Skip direction**) abandons the current direction and moves on, and Shift+Esc stops altogether.
3. The family of orbits connects HB1 to HB2 and turns at two folds of cycles: LPC1 at $i \approx 88.29$ and LPC2 at $i \approx 216.90$ (Figure 3). Between LPC1 and HB1 a stable orbit and the stable rest state coexist.

4. Click any point of a branch to jump there. This sets the parameter and loads that equilibrium, or exactly one period of that orbit, as the current state. Clicking a row of the special-points table does the same.
5. **Limit cycle** starts orbit continuation from the current trajectory instead. Integrate onto a stable cycle first (e.g. at $i = 150$), then press it.
6. Choose **period** (or **frequency**) under **Plot** to draw the period of the orbits against the parameter (Figure 5). The period grows from about 79 at HB1 to 135 at LPC1 and is about 42 at HB2. Sweeps show their periods too, where the oscillation is regular enough for one to be estimated.
7. Click a point on the orbit branch. As well as loading that orbit, this adds it to the **Special points** table as OR1, marked on the diagram with an open circle (Figure 4). Click another point to add OR2, and so on. Each picked orbit has a **Const. T** button, which follows it in two parameters at fixed period (Tutorial 3), and a **Remove** button.



SPECIAL POINTS 6

PT	I	V	PERIOD	
HB1 sub	93.682	−25.321	79.25	Follow Cycles
HB2 sub	212.06	7.8003	42.33	Follow Cycles
LPC1	88.293	23.925	135.4	Follow
LPC2	216.9	30.397	77.93	Follow
OR1	101.44	33.496	83.99	Const. T Remove
OR2	200.31	34.634	65.68	Const. T Remove

Figure 4: Left: two periodic orbits picked on the Morris–Lecar orbit branch, OR1 at $i \approx 100$ ($T \approx 84.0$) and OR2 at $i \approx 200$ ($T \approx 65.7$). Right: the **Special points** table with the picked orbits at the bottom, each with **Const. T** and **Remove**.

Sweep max/min provides a brute-force alternative. It integrates at a grid of parameter values, carrying the end state forward, and plots every local maximum and minimum after the transient. This is the right tool for chaotic or stochastic regimes, where continuation does not apply (Tutorial 4).

Sub- or supercritical? At a Hopf point with eigenvalues $\pm i\omega$, XPP Studio computes the first Lyapunov coefficient

$$\ell_1 = \frac{1}{2\omega}\,\mathrm{Re}\Big[\langle p, C(q,q,\bar{q})\rangle - 2\langle p, B(q, A^{-1}B(q,\bar{q}))\rangle + \langle p, B(\bar{q}, (2i\omega I - A)^{-1}B(q,q))\rangle\Big],$$

where A is the Jacobian, $Aq = i\omega q$, $A^{\mathsf{T}}p = -i\omega p$ and $\langle p, q\rangle = 1$ [5]. The multilinear forms B and C (the second and third derivatives of the vector field) are evaluated by finite differences. If $\ell_1 < 0$ the bifurcation is *supercritical*: a small stable orbit appears on the side where the equilibrium is unstable. If $\ell_1 > 0$ it is *subcritical*: an unstable orbit appears on the side where the equilibrium is stable, and the system usually jumps to a distant attractor.

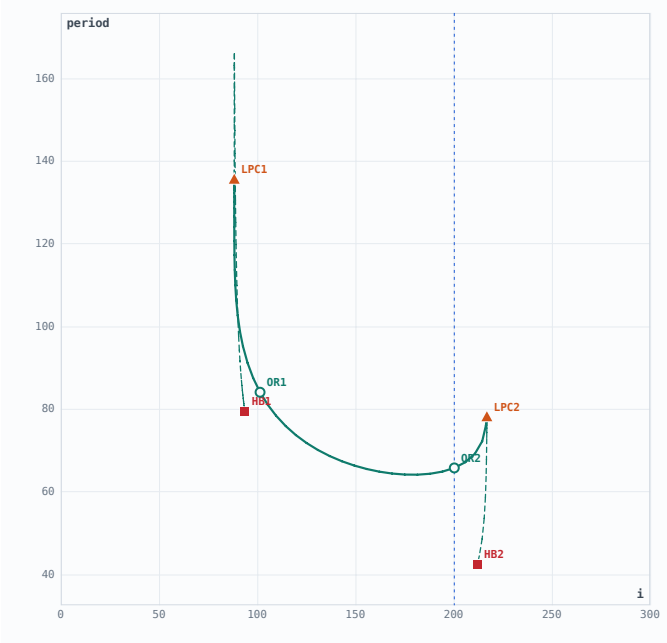


Figure 5: The same diagram with **Plot** set to **period**: the period of the periodic orbits against i, with the special points and the picked orbits.

SPECIAL POINTS 2

PT	I	V	PERIOD	
HB1 sub	93.682	−25.321	79.25	Follow Cycles
HB2 sub	212.06	7.8003	42.33	Follow Cycles

Figure 6: The **Special points** table after continuing equilibria of Morris–Lecar. Both Hopf points are tagged *sub* (subcritical). **Follow** continues a point in two parameters, and **Cycles** continues the periodic orbits born there.

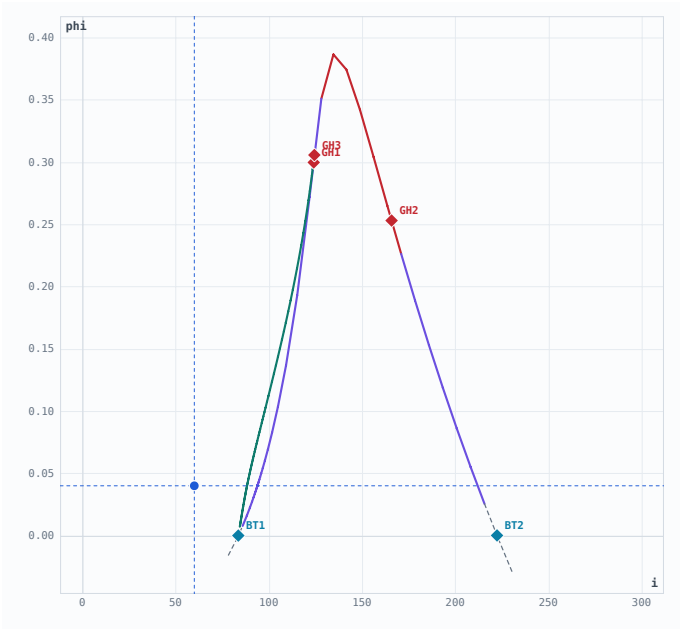
What is detected. For equilibria of flows: folds (LP), Hopf points (HB, tagged sub- or supercritical) and branch points (BP). For periodic orbits: folds of cycles (LPC), period doublings (PD) and torus (Neimark–Sacker) points (TR). For fixed points of maps: LP, PD and Neimark–Sacker (NS). At a PD of cycles, **Double** switches to the period-doubled branch.

6 Tutorial 3: two-parameter continuation

Any special point in the table with a **Follow** button can be continued in a second parameter. The Two-parameter view (5) shows the result, and its inspector panel selects the two parameters and their ranges.

Try it

1. After Tutorial 2, choose the second parameter (here ϕ, the rate of the recovery variable) in the Two-parameter panel.



TWO-PARAMETER CONTINUATION

Parameter 1 i

Range 0 to 300

Parameter 2 phi

Range 0 to 0.4

Start LPC1 · i 88.293

Follow curve | Clear curves

fold (LP) | Hopf, supercritical

Hopf, subcritical | neutral saddle

fold of cycles (LPC) | Default colours

CODIMENSION-TWO POINTS 5

PT	I	PHI	ON
BT1	83.618	2.7254e−5	HB
GH1	124.08	0.29974	HB
GH2	165.86	0.25305	HB
BT2	222.44	1.2279e−4	HB
GH3	124.42	0.30569	LPC

Figure 7: Left: two-parameter diagram for Morris–Lecar in (i, ϕ). A single Hopf curve joins HB1 and HB2. It is purple where the Hopf bifurcation is subcritical and red where it is supercritical, and the colour changes at the generalised Hopf (Bautin) points GH1 and GH2. Dashed ends are neutral saddles beyond the Bogdanov–Takens points BT1 and BT2 near $\phi = 0$. The green fold-of-cycles curve from LPC1 ends at GH3, which lies next to GH1. The blue cross-hairs mark the current parameter values. Right: the Two-parameter panel, with a colour picker for each kind of curve and the list of codimension-two points.

2. Set the range of ϕ to 0–0.4. Then press **Follow** on HB1 (in the Bifurcation view, or choose HB1 under **Start** and press **Follow curve**). Hopf curves are computed from a bordered system with the bialternate product [4, 5]. Here a single closed arch joins HB1 to HB2, so following HB2 as well would only retrace it.

3. The first Lyapunov coefficient ℓ_1 is computed at every point of the Hopf curve, and the curve is coloured by its sign: purple for subcritical, red for supercritical (Figure 7). Where ℓ_1 changes sign there is a *generalised Hopf* or *Bautin* point, labelled GH. For Morris–Lecar these are at about $(i, \phi) = (124.1, 0.300)$ and $(165.9, 0.253)$, so the Hopf bifurcations are supercritical only near the top of the arch. Bogdanov–Takens (BT), cusp (CP) and zero-Hopf (ZH) points are flagged in the same way. For maps, strong resonances R1 and R2 and fold–flip points are flagged too.

4. Press **Follow** on LPC1 to continue the fold of cycles. Its curve is computed by multiple shooting and is slower, typically 10–20 seconds. Bautin theory predicts that a curve of folds of cycles is born at every GH point, and here the LPC curve runs up the left side of the arch and ends at GH3 $\approx (124.4, 0.306)$. This is the endpoint of the orbit computation and agrees closely with GH1 from ℓ_1, an independent check of both.

5. Each kind of curve has a colour swatch in the panel legend: fold, sub- and supercritical Hopf, neutral saddle, fold of cycles, constant period and period doubling. Click a swatch to choose a new colour. The choice applies at once, is used by **Export SVG**, and is remembered in your browser. **Default colours** goes back to the theme colours.

6. Click a point on any curve to move to those parameter values. Then return to the phase plane or the bifurcation diagram to see the dynamics there.

7. **Curves of constant period.** Pick one or more orbits on the bifurcation diagram (Tutorial 2) and press **Const. T** on one of them, or choose it under **Start** as "OR1 (T=...)" and press **Follow curve**. The orbit is continued in both parameters with its period held fixed: the unknowns are the orbit, i and ϕ, and the extra equation is $T = T_0$. In Figure 8 the two gold curves are the parameter values where Morris–Lecar oscillates with period 84.0 and with period 65.7. Each ends where the orbit shrinks onto an equilibrium at a Hopf point whose period $2\pi/\omega$ equals T_0, so constant-period curves end on the Hopf curve. Integrating at points along a curve gives the same period to within the output spacing.

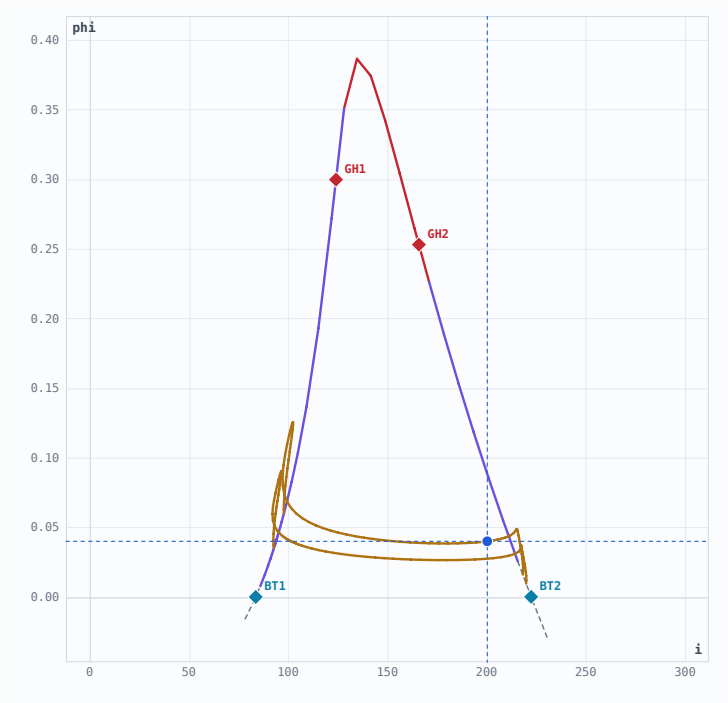


Figure 8: Curves of constant period (gold) for Morris–Lecar in (i, ϕ), traced from the picked orbits OR1 ($T \approx 84.0$) and OR2 ($T \approx 65.7$), together with the Hopf curve, coloured by criticality. Each constant-period curve ends on the Hopf curve, where the frequency of the Hopf point matches the fixed period.

Curves of period doublings of cycles (PDC) are continued the same way, which gives the classical period-doubling tongues of the Rössler system.

7 Tutorial 4: chaos in three dimensions

Load **Rössler system** [14] and press 2. Drag to rotate the attractor, scroll to zoom, or press **Spin**. The axes are chosen in the toolbar, and any of them can be time.

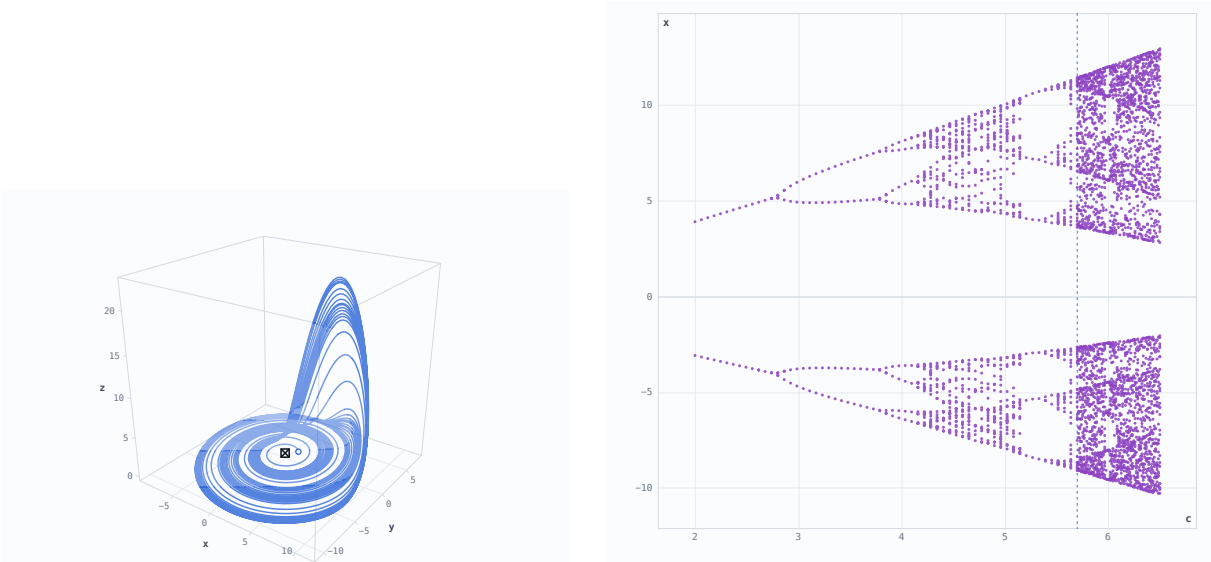


Figure 9: Rössler system at $a = b = 0.2$. Left: the chaotic attractor at $c = 5.7$ in the 3D view. Right: **Sweep max/min** in c, showing the local maxima and minima of x and the period-doubling route to chaos.

Try it

1. In the Bifurcation view choose c and press **Sweep max/min** (Figure 9, right). The number of parameter values is set by **Sweep samples** in the Numerics tab.
2. For a continuation version, set $c = 2.5$, integrate onto the stable cycle and press **Limit cycle**. Period doublings are found at $c \approx 2.8324$, 3.8374, 4.1242 and 4.1870. Pressing **Double** at each in turn follows the cascade.
3. **Follow** on a PD point traces its curve in (c, a).

8 Tutorial 5: maps

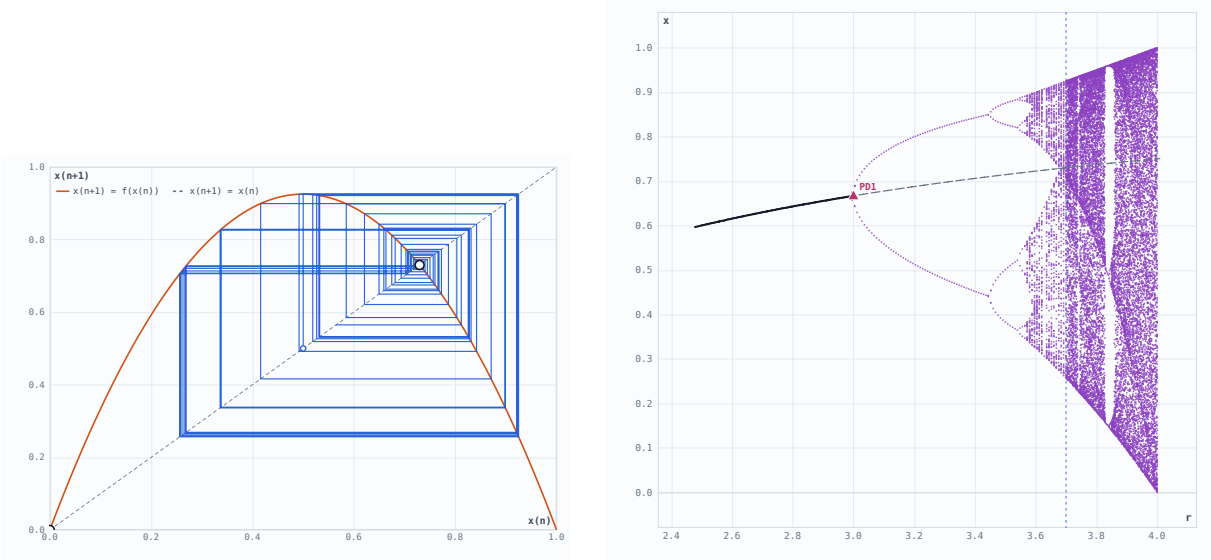


Figure 10: Logistic map $x \mapsto rx(1 - x)$. Left: cobweb diagram at $r = 3.7$. Right: orbit diagram from **Sweep max/min** (purple dots), with the continued fixed point and its period doubling PD1 at $r = 3$.

A model is a map when its equations are written `x(t+1)=...`. Trajectories are then drawn as points, and for a one-variable map the phase plane becomes a cobweb diagram (Figure 10). For maps, **Sweep max/min** keeps up to 250 iterates after the transient, which gives the familiar orbit diagram. Continuation finds folds, period doublings and Neimark–Sacker points of fixed points. The **Hénon map** example shows a fold at $a \approx -0.1225$ and a PD at $a \approx 0.3675$, and the **Delayed logistic map** an NS bifurcation at $a = 2$.

9 Tutorial 6: noise and random switching

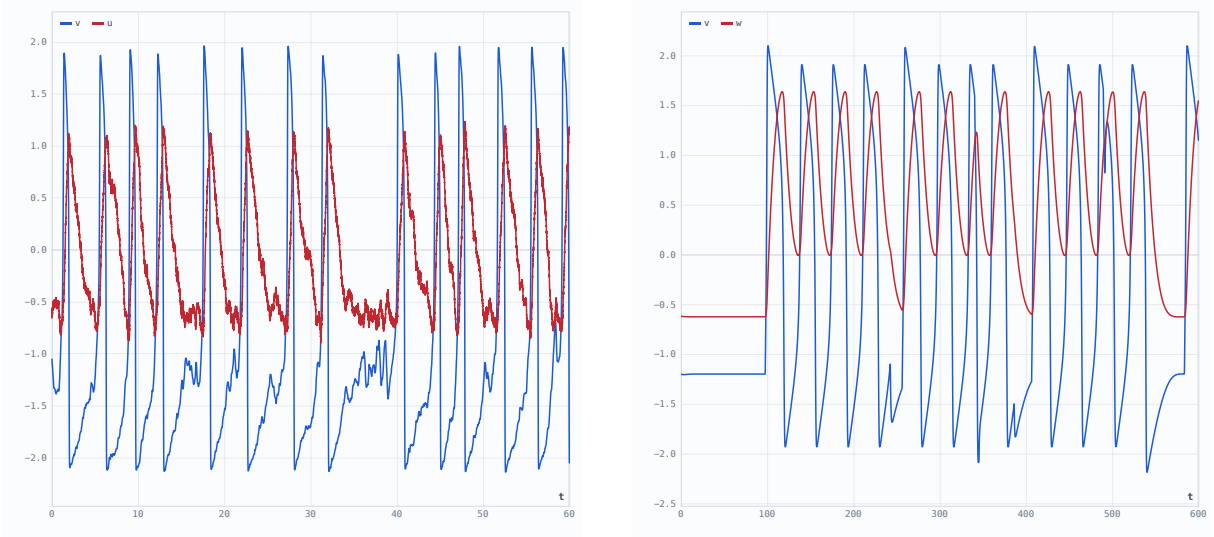


Figure 11: Left: excitable FitzHugh–Nagumo driven by white noise; noise triggers irregular spikes. Right: FitzHugh–Nagumo whose input is a two-state Markov variable, so oscillation switches on and off at random.

Wiener noise. Declaring `wiener w` makes `w` a white-noise source. The equations are integrated by Euler–Maruyama with step `dt`, and over each step `w` is $N(0,1)/\sqrt{dt}$, exactly as in XPPAUT. In **Noisy FitzHugh–Nagumo** the rest state is stable, and noise of amplitude `sig` kicks the system over threshold (Figure 11, left). Drag `sig` to change the firing rate.

Markov variables. **Telegraph-driven FitzHugh–Nagumo** declares

```
markov s 2
{0} {kon}
{koff} {0}
```

a variable that jumps $0 \to 1$ at rate `kon` and $1 \to 0$ at rate `koff`. Rates may depend on the state variables. The input `i0+di*s` switches the cell between rest and oscillation (Figure 11, right).

Seeds. Each stochastic run uses a new random seed, which is reported in the status line. Enter a number in **Noise seed** (Numerics tab), or write `@ seed=7`, to repeat the same noise path, for instance to compare parameter values. Equilibria, nullclines and continuation always use the noise-free system, with Markov states held at the values in the Initial tab.

10 Tutorial 7: stiff systems

The **Stiff van der Pol** example has $\mu = 1000$: relaxation oscillations whose slow and fast time scales differ by a factor of about 10^6. With `@ meth=stiff` the Rosenbrock 2(3) solver integrates a period ($T \approx 1614.4$) in a few thousand steps, where an explicit method needs hundreds of thousands. When the stiff method is selected, periodic-orbit continuation also uses it for its shooting segments, so the period can be continued down in μ.

11 Tutorial 8: arrays and Volterra equations

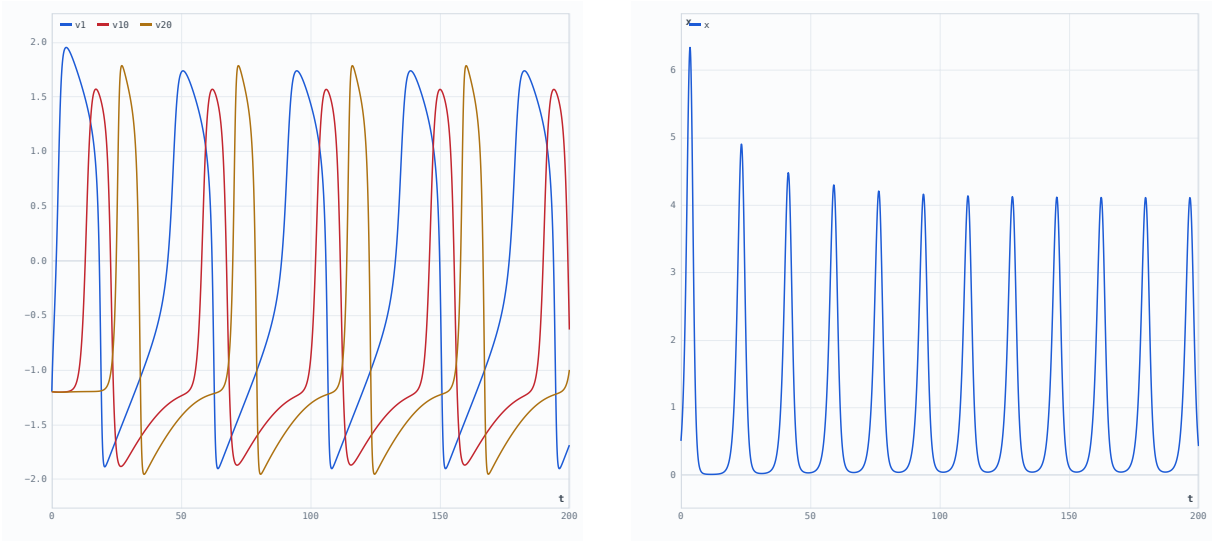


Figure 12: Left: a chain of 20 FitzHugh–Nagumo cells written with arrays. Spikes start at cell 1 and arrive later at cells 10 and 20. Right: logistic growth with a gamma-distributed delay, written as a Volterra convolution. At $r\tau = 2.5$ the equilibrium is unstable and the population oscillates.

Arrays. XPPAUT's arrays are a text macro. The **Chain of 20 FitzHugh–Nagumo cells** example is written as

```
%[1..20]
v[j]'=v[j]-v[j]^3/3-w[j]+i1*heav(1.5-[j])+d*(v[j+1]-2*v[j]+v[j-1])
%
w[1..20]'=eps*(v[j]+a-b*w[j])
v0=v1
v21=v20
init v[1..20]=-1.2
init w[1..20]=-0.62
```

The block between `%[1..20]` and `%` is repeated for $j = 1, \ldots, 20$. Every `[expr]` is replaced by its integer value, so `v[j-1]` becomes `v3` when $j = 4$, and `[j]` on its own becomes the number j. A single line containing `[1..20]` is repeated the same way. The load message reports how many lines the file expanded to. The expanded model is an ordinary 40-dimensional system, so its equilibria, eigenvalues (computed by the QR algorithm) and continuation all work as usual. In Figure 12 (left) the time course shows v_1, v_{10} and v_{20}; choose them with the chips in the inspector.

Array plots. For a chain or lattice it is often clearer to see all the cells at once, as in XPPAUT's array plot. The strip below the phase plane has a **Time series** | **Array** switch at its top right. In

array mode each ticked variable is a row and time runs across, and each point is coloured by its value (Figure 13): travelling pulses in the chain appear as diagonal stripes, and their slope is the propagation speed.

- The **Initial** tab has an **Array plot** section with a tick box for every variable, and **All**, **None** and **Show** buttons. The rows appear in the order of the variables, and by default all state variables are ticked. For the chain, press **None** and tick $v_1, \ldots, v_{20}$.
- **Colours...** opens a small dialogue with the colours for the minimum and maximum values and, optionally, the values themselves; left empty, they are the smallest and largest values of the ticked variables along the trajectory. Colours in between blend evenly (in the perceptual OKLab space), and the colour bar on the right of the strip shows the range.

Volterra integrals. XPP Studio supports XPPAUT's integral terms:

- `int{K(t,t',x)}` $= \int_0^t K(t, t', x(t'))\, dt'$, where the variables inside are evaluated at t';
- `int{K#x}` $= \int_0^t K(t - t')\, x(t')\, dt'$, a convolution with the kernel written as a function of t;
- `int[mu]{...}` adds the weakly singular factor $(t - t')^{-\mu}$, $0 \le \mu < 1$;
- `u(t)=...` or `volterra u=...` defines a variable by an integral equation.

The **Logistic growth with distributed delay** example is

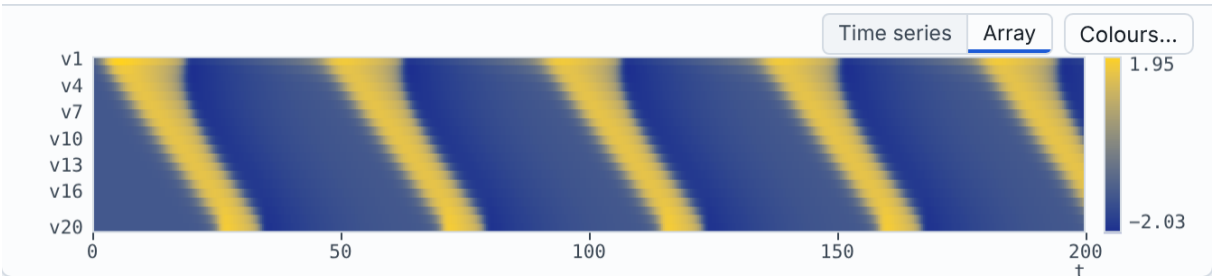
```
x'=r*x*(1-int{t*exp(-t/tau)/tau^2#x})
par r=1,tau=2.5
@ total=200,dt=0.05,vmaxpts=1000
```

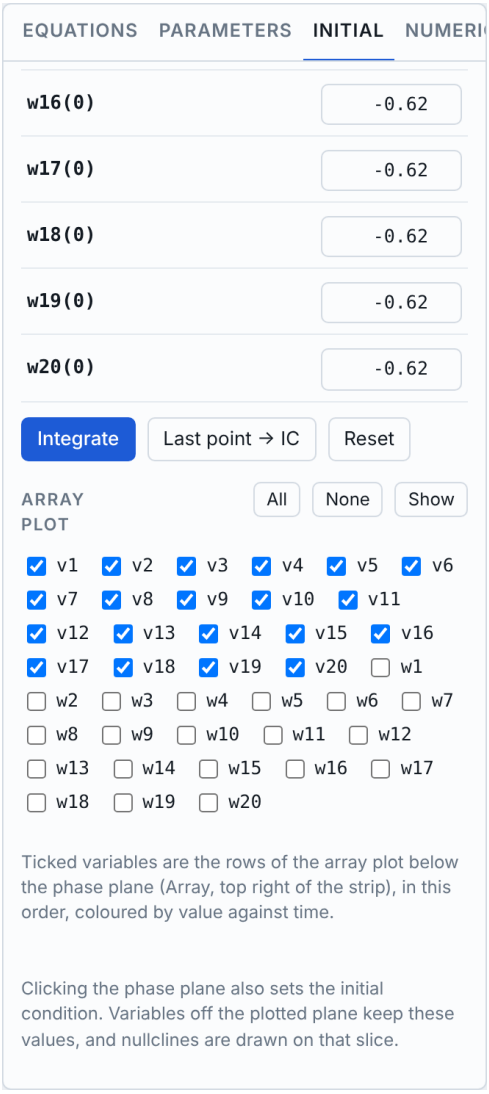
The kernel $K(s) = s\, e^{-s/\tau}/\tau^2$ has mean delay 2τ. Linearising about $x = 1$ gives $\lambda(1+\lambda\tau)^2 + r = 0$, so the equilibrium loses stability in a Hopf bifurcation at $r\tau = 2$. Sweeping τ with **Sweep max/min** shows the oscillation appear.

Integro-differential equations are solved by a fixed-step method: the trapezoid rule for the differential part and product-trapezoid quadrature [8] for the integrals. The quadrature weights are exact for the $(t - t')^{-\mu}$ factor, and the implicit terms are found by fixed-point iteration. The error falls as dt^2, or as $dt^{2-\mu}$ with a singular kernel. The method reproduces $u(t) = e^t$ for $u = 1 + \int_0^t u$, and the solution $e^{\pi t}\,\mathrm{erfc}(\sqrt{\pi t})$ of the Abel equation `u(t)=1-int[.5]{1#u}`. It also agrees with the equivalent three-variable ODE for the gamma kernel. Only the last `vmaxpts` steps are remembered (default 4000). Because the state of such a system is its whole history, equilibria, nullclines and continuation are switched off for these models. Trajectories and sweeps work normally.

Periodic variables and tables. Two further XPPAUT features are worth knowing:

- `@ fold=th` keeps an angle in $[0, 2\pi)$ (or $[0,$ `tor_per`$)$). Plots break the trajectory where it wraps, and continuation measures distances modulo the period; see the **Driven pendulum** example.
- Tables (**Phase oscillators with a tabulated H**) are interpolated linearly. They can be sampled from an expression or read from a data file.





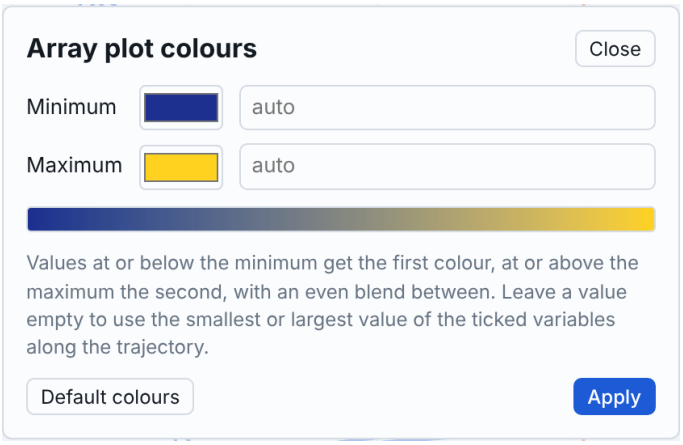


Figure 13: Top: the array plot of $v_1, \ldots, v_{20}$ for the chain of FitzHugh–Nagumo cells. Each pulse starts at cell 1 and crosses the chain in about 20 time units. Bottom left: choosing the rows in the Initial tab. Bottom right: the colour dialogue.

12 Tutorial 9: delay equations

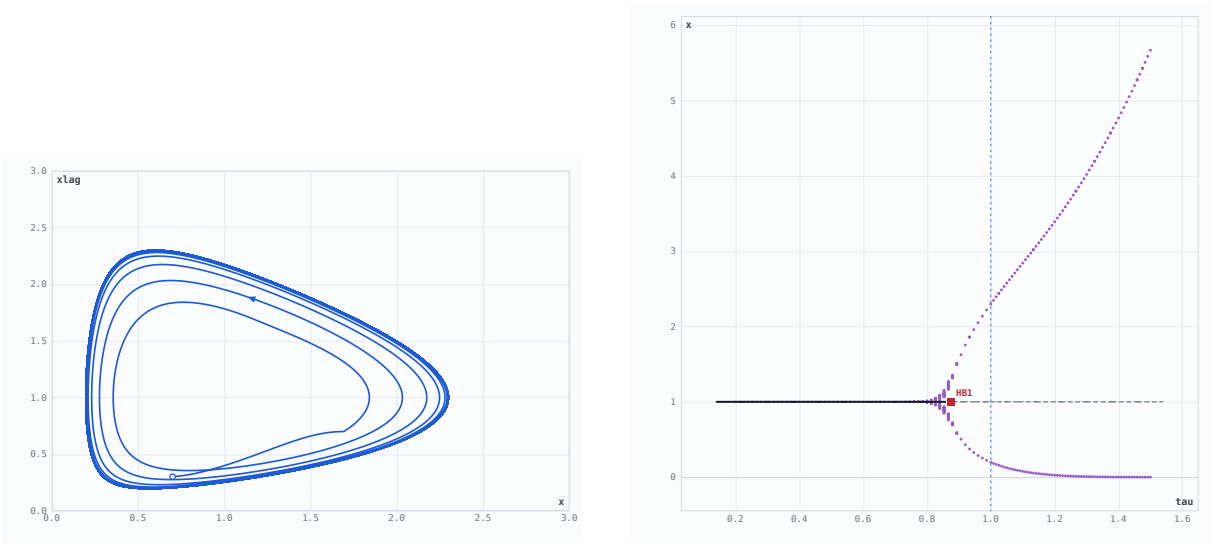


Figure 14: Hutchinson's delayed logistic equation with $r = 1.8$. Left: $x(t-\tau)$ against $x(t)$ for $\tau = 1$, starting from the history $0.5 + 0.2\cos 3t$ and settling onto a periodic orbit. Right: bifurcation diagram in τ. The equilibrium $x = 1$ (black) loses stability at the Hopf point HB1, $\tau \approx 0.874$, close to the exact value $\pi/(2r) = 0.873$. The purple dots, from **Sweep max/min**, show the oscillation that grows from it.

A delay-differential equation uses past values of its variables. In XPP Studio, as in XPPAUT, `delay(x,tau)` stands for $x(t-\tau)$, where `x` must be a variable with an equation and `tau` is any expression; it may depend on the state. Load **Delayed logistic equation**, Hutchinson's model [12]:

```
x'=r*x*(1-delay(x,tau))
aux xlag=delay(x,tau)
par r=1.8,tau=1
x(0)=0.5+0.2*cos(3*t)
```

History. A delay equation needs its state on a whole interval $[-\tau, 0]$, not only at $t = 0$. Writing `x(0)` as a formula in `t` gives that history, and its value at $t = 0$ becomes the initial condition. Without a formula, the history is held constant at the initial value. The formulas also appear in the **Initial** tab under **History for t < 0** (Figure 15), where they can be changed or cleared. Clicking the phase plane changes only the values at $t = 0$. The auxiliary variable `xlag` makes $x(t-\tau)$ available for plotting, so the phase plane in Figure 14 shows $x(t-\tau)$ against $x(t)$, a common way to draw the attractor of a scalar delay equation.

Integration. Delay equations run forward from $t = 0$ with fixed-step fourth-order Runge–Kutta (adaptive methods fall back to it). Every step is stored, and delayed values are found by cubic Hermite interpolation of the stored solution, or from the history when $t - \tau < 0$. For $x' = -x(t-1)$ with $x = 1$ on $[-1, 0]$, whose exact solution is piecewise polynomial, the computed $x(2)$ and $x(3)$ agree with the exact values to rounding error.

Stability and continuation. Equilibria of a delay equation are those of the system with the delays removed, but their stability is not. It is set by the roots of the characteristic equation

$$\det\Big(\lambda I - A_0 - \sum_k B_k\, e^{-\lambda \tau_k}\Big) = 0,$$

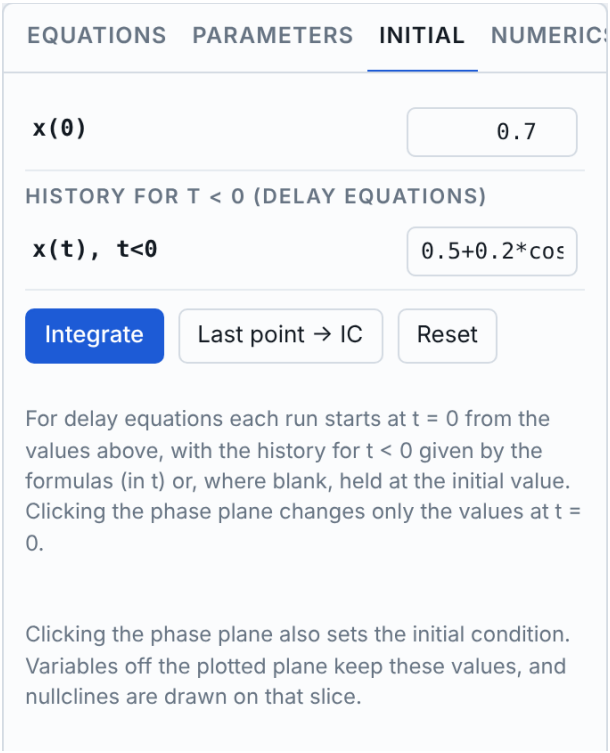


Figure 15: The Initial tab for a delay equation: the value at $t = 0$ and, below it, the history for $t < 0$ as a formula in t.

where A_0 and B_k are the derivatives with respect to the current and the delayed values. There are infinitely many roots, but only finitely many lie to the right of any vertical line. XPP Studio approximates the rightmost ones by Chebyshev collocation of the solution operator [11] and then polishes them with Newton's method. The equilibrium list then reports, for example, *Unstable, 2 roots with Re* $\lambda > 0$ and lists the rightmost roots. **Continue equilibria** uses the same roots, so Hopf points are those of the delay equation.

Try it

1. Load **Delayed logistic equation** and press G. With $r\tau = 1.8 > \pi/2$ the solution settles onto a periodic orbit (Figure 14, left). Change the history formula in the Initial tab and integrate again.
2. The inspector lists two equilibria. $x = 0$ has the single root $\lambda = r$. $x = 1$ has a complex pair $\lambda \approx 0.097 \pm 1.63i$ in the right half-plane.
3. In the Bifurcation view choose `tau`, a range of 0.2 to 1.5 and **Step ds** 0.02, then press **Continue equilibria**. A Hopf point HB1 is found at $\tau \approx 0.874$, against $\pi/(2r) = 0.8727$. **Sweep max/min** adds the oscillation amplitude (Figure 14, right).
4. Load **Mackey–Glass equation** [13] (Figure 16). With $\tau = 17$ the solution is chaotic; the file uses `@ trans=300` to drop the transient. Continuation in τ finds Hopf points at $\tau \approx 4.71$ and 20.98. Linearising by hand gives $\lambda = -0.1 - 0.4\,e^{-\lambda\tau}$, with Hopf points at $\tau = 4.708$ and 20.93.

Periodic-orbit and two-parameter continuation, the direction field and the derivative readout are switched off for delay equations, because the state at time t is a whole function rather than a point.

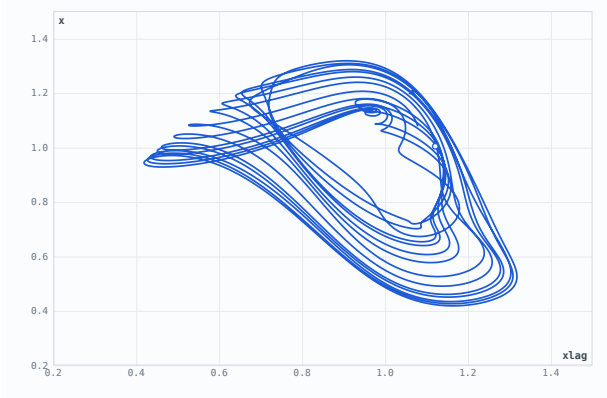


Figure 16: The chaotic Mackey–Glass attractor at $\tau = 17$, drawn as $x(t)$ against $x(t-\tau)$ after a transient of 300.

13 Tutorial 10: events and global flags

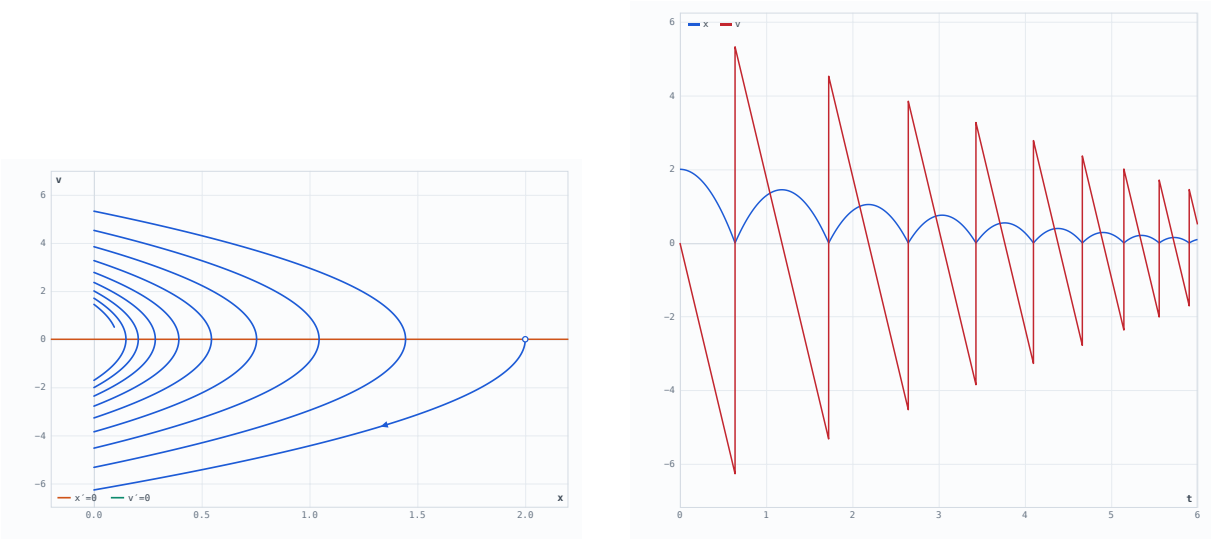


Figure 17: The bouncing ball. Left: velocity against height. Each parabola is one flight, and the line breaks at each event instead of joining the points before and after it. Right: height x (blue) and velocity v (red) against time, with the jump in v at each bounce.

Many models change discontinuously when a threshold is reached: a neuron fires and resets, a ball bounces, a switch flips. XPPAUT describes such events with *global flags*,

```
global sign condition {name1=expr1;name2=expr2;...}
```

The assignments are made when `condition` crosses zero from negative to positive (`sign`$= 1$), from positive to negative (-1), or in either direction (0). All the right-hand sides are evaluated first, then all the assignments are made together. The names may be state or Markov variables, and `arret=1` stops the integration.

How events are located. After every step XPP Studio checks each condition. If one has crossed zero in its direction, the crossing time is found on the cubic Hermite interpolant of that step, by regula falsi on the condition. The point just before the event and the point just after the assignments are both stored, at the same time, and integration continues from there. This works with every integrator, including the adaptive and stiff ones, and with maps, where the assignments are applied to the new iterate.

Try it

1. Load **Bouncing ball**:

   ```
   x'=v
   v'=-g
   global -1 x {x=0;v=-e*v}
   ```

   Press G. The status line reports the number of events. The first two bounces occur at $t = \sqrt{2x_0/g} = 0.6386$ and $t = 0.6386\,(1 + 2e) = 1.7242$, and the computed times agree to six decimal places. Each flight is a parabola in the phase plane (Figure 17). Lower the restitution e with its slider and watch the bounces die away faster.

2. Load **Integrate-and-fire neuron**:

   ```
   v'=(-v+i-w)/tau
   w'=-w/tauw
   global 1 v-1 {v=0;w=w+b}
   ```

   Each time v reaches threshold it is reset and the adaptation variable w is kicked up, so the firing rate slows until it settles (Figure 18). With $b = 0$ the interspike interval is $\tau \ln(i/(i-1))$, and the computed value agrees to six or more digits.

3. Add `global 1 t-50 {arret=1}` to a model to stop every run at $t = 50$.

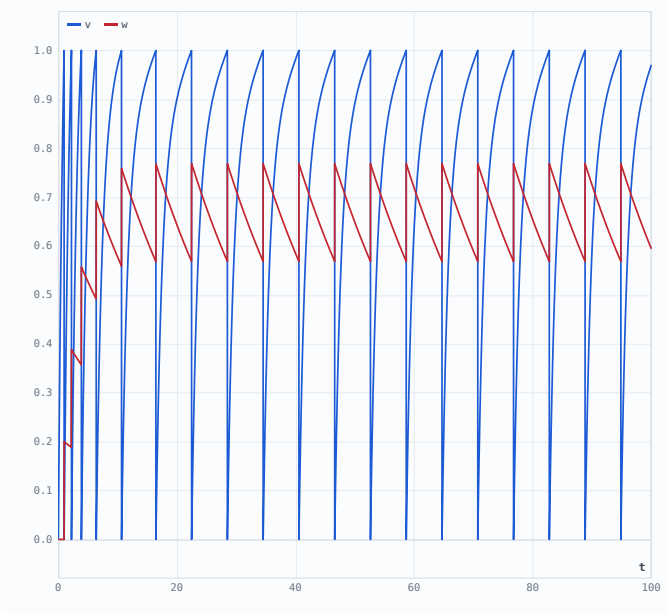


Figure 18: Integrate-and-fire neuron with adaptation: v (blue) is reset to 0 at each threshold crossing, while the adaptation current w (red) builds up and lengthens the interspike interval.

Equilibria, nullclines and equilibrium continuation ignore flags. Periodic-orbit continuation is switched off for models with flags; use **Sweep max/min** for the long-term behaviour instead.

14 Numerical settings

The **Numerics** tab (Figure 19) collects the integration and continuation parameters. Changes to the integration settings rerun every trajectory on screen.

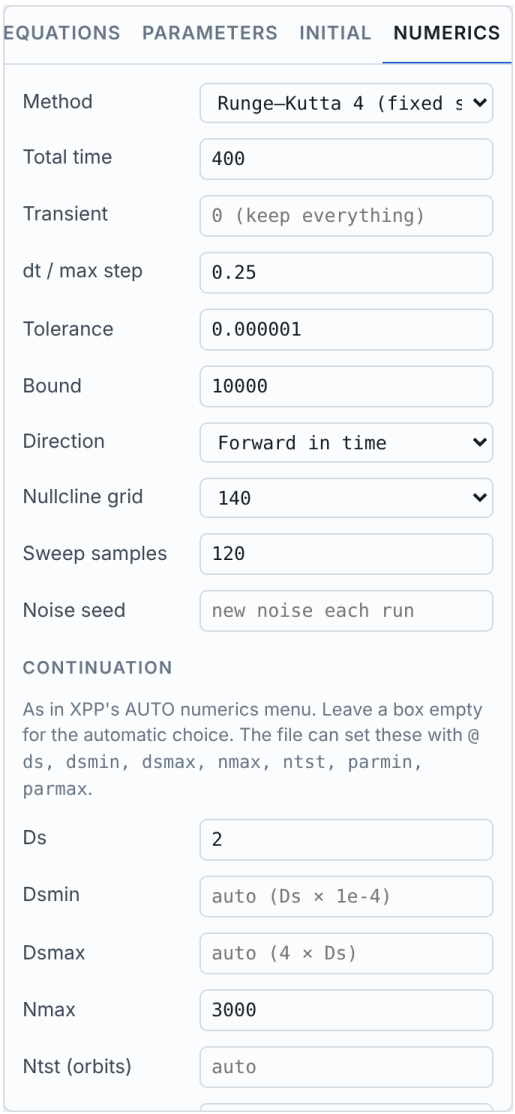


Figure 19: The Numerics tab: integration settings at the top, continuation settings below.

Integration.

Method As in Section 3.

Total time The duration of each run; for maps, the number of iterations.

Transient As in XPPAUT, the run still lasts the total time, but only $t \geq$ the transient is kept (`@ trans=`). Use it to look at attractors without the approach to them. Manifolds and the single periods loaded from continuation are never cut.

dt The step for fixed-step methods, and the largest step for adaptive ones.

Tolerance Applies to the adaptive methods.

Bound Integration stops when any variable exceeds this value.

Direction Forward or backward in time.

Nullcline grid Sets the resolution of the nullclines.

Sweep samples Sets the number of parameter values in **Sweep max/min**.

Continuation. These settings correspond to the entries of XPPAUT's AUTO numerics window and can be set in the file with `@ ds, dsmin, dsmax, nmax, ntst, parmin, parmax`. Leave a box empty to use the automatic choice.

Setting	Effect
Ds	initial arclength step for equilibrium branches (same as **Step ds**)
Dsmin, Dsmax	smallest and largest steps; a branch stops if the step falls below Dsmin
Nmax	maximum number of points in each direction of an equilibrium branch
Ntst (orbits)	number of shooting intervals per periodic orbit (automatic: 20 for up to 3 variables, 14 up to 5, otherwise 10)
Ds, Dsmax, Nmax (orbits)	the same for periodic-orbit branches and curves of cycles
Nmax (2-par), Nmax (2-par cycles)	point limits for two-parameter curves

If a branch stops early, the status line says why: for example, it reached the point limit, the corrector stopped converging, the orbit shrank onto a Hopf point, or the period grew without bound (a homoclinic orbit is near). If it stopped because of a step-size or point limit, adjust the corresponding setting and run it again.

15 Saving and exporting results

Export SVG Saves the current view as a vector figure with light-theme colours, ready for a paper or for editing in Inkscape or Illustrator. All the plots in this document were made this way.

Export CSV Available in the Time course, Bifurcation and Two-parameter views.

- The time-course table has every trajectory on screen, one row per stored point, with the trajectory number, its kind (trajectory or manifold), its direction (forward or backward), t and every variable.
- The bifurcation table has one row per computed point, with these columns: curve, kind (equilibrium, periodic orbit, sweep or special), label (HB1, LPC2, ...), parameter name and value, stability, number of unstable directions, period, the state, and the maximum and minimum of each variable along an orbit.
- The two-parameter table has the two parameter values, the period where relevant, the state, and labelled rows for codimension-two points. Hopf curves also get columns for ℓ_1 and for criticality (subcritical, supercritical or neutral saddle).
- In the bifurcation table, special rows for Hopf points give ℓ_1, and the stability column says subcritical or supercritical.

Export .dat In the Bifurcation and Two-parameter views, saves the same continuation data as `.dat` files: numbers only, no header and space-separated columns, the form XPPAUT and gnuplot read. The bifurcation diagram gives one file each for equilibria, periodic orbits and sweeps. The status line lists the columns, and text fields become numeric codes.

Copy CSV, Save .dat In the trajectory list, copy the current trajectory (time, variables and auxiliaries) to the clipboard, or save it as a `.dat` file.

Copy text Copies the `.ode` file, including any edits, so that it can be saved or used in XPPAUT.

Writing the current state into the file

After exploring a model you will often want to keep the parameter values and settings you have arrived at. **Write state**, in the Equations tab, writes them into the `.ode` text, much as XPPAUT

does when it saves a file:

- parameter values and initial conditions are changed where they are defined, in the `par`, `init` and `x(0)=` lines, as are slider ranges in `#@range` lines and history formulas of delay equations;
- the integration settings (method, total time, step, transient, tolerance, bound, random seed), the plot axes and window, and the continuation settings are collected in `@` lines just before `done`.

For example, after setting $i = 150$ and a transient of 100 in the Morris–Lecar example, the end of the file reads

```
param i=150,c=20
...
# Settings written by XPP Studio, 2026-10-20 10:30
@ meth=rungekutta, total=400, dt=0.25, trans=100, bound=10000
@ xp=v, yp=w, zp=ica, xlo=-75, xhi=45, ylo=-0.1, yhi=0.55
@ ds=2, nmax=3000, parmin=0, parmax=300
done
```

Pressing **Write state** again replaces this block rather than adding another, and **Apply** on the result restores exactly the same state. The text is not reloaded, so the trajectories on screen are kept. Use **Copy text**, or your browser's own storage of the editor, to keep the file. Parameters defined through array lines are left as written.

16 Differences from XPPAUT

XPP Studio aims to read the same files and to feel familiar, but it is a reimplementation:

- **Continuation.** It is built in and written from scratch: pseudo-arclength continuation of equilibria, and multiple shooting (rather than AUTO's orthogonal collocation) for periodic orbits. Results agree with AUTO to plotting accuracy on the examples, but very long or very unstable orbits may need more shooting intervals (Ntst).
- **Delay equations** are integrated with a fixed step only, and periodic orbits of delay equations cannot be continued. XPPAUT's other delay options, such as `delshft` and `ndelay`, are not supported. The `@ delay` option is accepted but not needed, since all of the history is kept.
- **Global flags** can set state and Markov variables and `arret`. The XPP output flag `out_put` is accepted and ignored.
- **Not supported:** boundary-value problems (`bdry`), `special` and `export` declarations, and `set` blocks.
- **Two-parameter Hopf curves** are limited to 12 variables, because the bialternate product grows quadratically with dimension.
- **Volterra models** cannot be continued, since their state is infinite-dimensional.
- **Browser files.** Files are opened and saved through the browser. Table data files are kept in the browser's local storage until you press **Forget table files**.

17 Credits and licence

XPP Studio is inspired by, and reads the model files of, XPPAUT by G. Bard Ermentrout (University of Pittsburgh), described in [1]. Its continuation methods follow the ideas of AUTO

[2] by Eusebius Doedel and colleagues, and of MatCont [3]. This version (October 2026) was created by Peter Ashwin and Bard Ermentrout in collaboration with Claude Opus 5.5, an AI model made by Anthropic.

XPP Studio is free software, released under the GNU General Public License, version 3 or later. The licence text is distributed with the program (`LICENSE.txt`) and is available at `https://www.gnu.org/licenses/gpl-3.0.html`.

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